

TREE-01

Q1)

Question:

DS

Session

Tree 1

Question Information

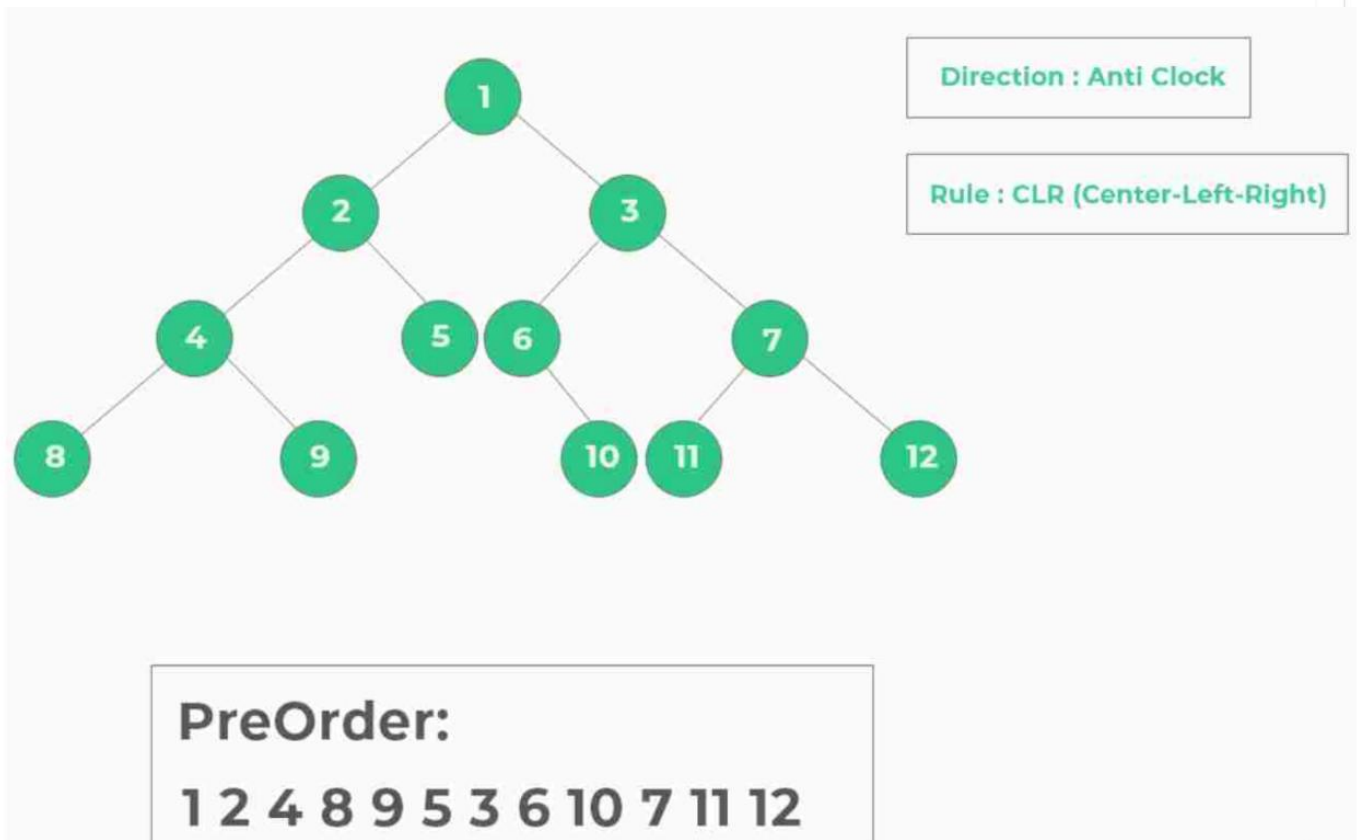
Level 1

Challenge 61

Question description

Jegan Sir students were chatting and playing quite loudly on the last day of the year, celebrating the end of the academic session. Jegan sir was harshly chastised by the college's principal. But, instead of becoming enraged, he attempted to engage everyone in a different task.

So Jegan sir gave his students to solve the task such that, you have to perform pre-order tree traversal in Binary search tree.



Constraints:

0 < size < 100

0 < data < 1000

PROGRAM:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *left, *right;
7  };
8
9  struct Node* create(int x) {
10     struct Node* n = malloc(sizeof(struct Node));
11     n->data = x;
12     n->left = n->right = NULL;
13     return n;
14 }
15
16 struct Node* insert(struct Node* root, int x) {
17     if (root == NULL)
18         return create(x);
19
20     if (x < root->data)
21         root->left = insert(root->left, x);
22     else
23         root->right = insert(root->right, x);
24
25     return root;
26 }
27
28 void preorder(struct Node* root) {
29     if (root == NULL)
30         return;
31
32     printf("%d ", root->data);
33     preorder(root->left);
34     preorder(root->right);
35 }
36
37 int main() {
38     int n, x, i;
39     struct Node* root = NULL;
40
41     scanf("%d", &n);
42
43     for (i = 0; i < n; i++) {
44         scanf("%d", &x);
45         root = insert(root, x);
46     }
47
48     preorder(root);
49
50     return 0;
51 }
```

OUTPUT:

TEST CASE 1:

```
7
200 120 80 160 280 240 320
200 120 80 160 280 240 320
```

TEST CASE 2:

```
9
34 67 10 90 410 810 40 20 60
34 10 20 67 40 60 90 410 810
```